Homework 2: What Is the Target?

Statistical Evaluation of AI · Spring 2027

Out / due	Tue Jan 26 / Tue Feb 2, 11:59pm
Points	100: Problem 1 (50), Problem 2 (30), Problem 3 (20)
Expected time	About 5 hours after completing Lab 2; contact course staff if setup/debugging alone exceeds 30 minutes
Reading	Plank, paper pp. 1–5; DynaSent Figure 1 and §§3.3–3.4, 4.2–4.5; Shalizi, "Some basic math about averaging" and "Exchangeable variables and random limits for averages"; the short Statistical Proof Book proof of the law of total covariance; Bishop & Bishop Chapter 5 and Exercise 5.1 as technical support
AI policy	No AI for Problem 1. AI is optional in Problem 2(a–d) and required for the 5-point attempt in Problem 2(e) and for Problem 3. Submit the AI-use record at the end.
Working file	Complete all work in `hw02_starter.ipynb`; the PDF is a read-only prompt copy.
Submit	Notebook PDF + completed `.ipynb` in one Gradescope submission.

Connection to Lab 2

Lecture introduced labels as measurements produced by a protocol. Lab 2 used a DynaSent teaching subset to inspect disagreement, compare construction rounds, and construct hard and soft outcomes from five rater labels. This homework asks you to explain the theory behind that construction, examine what aggregation and item selection hide, and evaluate forecasts against targets made from new, disjoint DynaSent measurements.

The throughline: one sentence, five different objects

The central question is **what, exactly, is the target?** Keep these objects separate throughout the homework:

Object	Role in this homework
L_i	one rater's categorical measurement of a sentence
q	the response distribution induced by the working population and labeling model
$\hat{q}$	the empirical soft target constructed from the observed rater measurements
Y_{maj}	the hard target constructed by majority vote
p	a forecaster's probability report, made before observing the measurements

The working model does not declare one observed label to be revealed truth. Instead, it lets us ask what repeated labels estimate, what majority vote discards, and how a constructed target shapes model evaluation.

You may reuse the lab's rater-expansion, vote-summary, Brier-score, and denominator-first table code. The homework uses different items and asks for a new statistical argument. As self-checks, every item must have exactly five votes, the three rater proportions must sum to one, and every unnormalized three-class Brier score must lie between 0 and 2.

Problem 1 — A label is a random outcome (50 points)

Complete this problem without AI assistance. Show the mathematical steps that support each conclusion; isolated numerical answers receive little credit. Parts (a)(i)–(iii) begin with Bishop & Bishop Exercise 5.1 and extend it to the Brier score used in lecture and lab.

Notation and working model. Fix a sentence x. Let L be a random label generated according to the working model

$$L \mid x \sim \text{Categorical}(q_{\text{neg}}, q_{\text{neu}}, q_{\text{pos}}).$$

Thus $q_k = P(L = k \mid x)$ is the conditional probability of label k under the working model, and the three probabilities are nonnegative and sum to one. For each class k, define the indicator

$$T_k = \mathbf{1}\{L = k\},$$

which equals one when $L = k$ and zero otherwise. The vector $T = (T_{\text{neg}}, T_{\text{neu}}, T_{\text{pos}})$ is the one-hot encoding of L: exactly one component equals one. For example, a positive label gives $T = (0, 0, 1)$. Suppose we must choose a probability vector $p = (p_{\text{neg}}, p_{\text{neu}}, p_{\text{pos}})$ to report before observing L. In this derivation, p is a fixed candidate report; the expectation is over the random label $L \mid x$. The three-class Brier score is

$$B(p, T) = \sum_{k=1}^{3} (p_k - T_k)^2.$$

Each numbered item is a separate grading checkpoint. Show your setup and intended next step; correct partial work can earn substantial credit.

a. **One categorical label and its expected Brier score (20 points).**

(i) Categorical label and indicators (4 points). Write the three possible values of L and the corresponding one-hot vectors T. Explain why the three indicators cannot be treated as independent Bernoulli random variables even though each indicator is zero or one.

(ii) Expected value of an indicator (4 points). Use the definition of expected value for a zero-one variable to complete

$$\begin{aligned} E[T_k \mid x] &= 1 \cdot P(T_k = 1 \mid x) + 0 \cdot P(T_k = 0 \mid x) \\ &= \underline{\qquad\qquad\qquad\qquad}. \end{aligned}$$

Then use $T_k = \mathbf{1}\{L = k\}$ to show that $E[T_k \mid x] = P(L = k \mid x) = q_k$.

(iii) One-class squared loss (6 points). Hold p_k fixed and show that

$$E[(p_k - T_k)^2 \mid x] = (p_k - q_k)^2 + q_k(1 - q_k).$$

(iv) Sum, interpret, and minimize (6 points). Show that summing part (iii) over the three classes gives

$$E[B(p, T) \mid x] = \sum_{k=1}^{3} (p_k - q_k)^2 + \sum_{k=1}^{3} q_k(1 - q_k).$$

In 1–2 sentences, explain what each of the two sums represents and identify which one depends on the candidate report p. Find the value of p that minimizes the expected Brier score under this model, and explain what your answer means in context.

b. **Repeated labels estimate class probabilities (15 points).** For the same sentence x, suppose we observe m conditionally independent labels $L_1, \dots, L_m$ from the working categorical model with common probability vector q. Define

$$T_{ik} = \mathbf{1}\{L_i = k\}, \qquad N_k = \sum_{i=1}^{m} T_{ik}, \qquad \hat{q}_k = \frac{N_k}{m}.$$

(i) Structural identities (3 points). Show that $\sum_{k=1}^{3} N_k = m$ and $\sum_{k=1}^{3} \hat{q}_k = 1$.

(ii) Mean and variance (6 points). Use indicator expectations and the conditional independence assumption to show that

$$E[\hat{q}_k \mid x] = q_k, \qquad \mathrm{Var}(\hat{q}_k \mid x) = \frac{q_k(1-q_k)}{m}.$$

(iii) Dependence across classes (4 points). For $j \neq k$, use the fact that $T_{ij}T_{ik} = 0$ to show that

$$\mathrm{Cov}(\hat{q}_j, \hat{q}_k \mid x) = -\frac{q_j q_k}{m}.$$

In one sentence, explain why the covariance is negative.

(iv) More labels (2 points). In one sentence, explain what changes—and what does not change—as m increases.

c. **Expected Brier score for an empirical soft target (15 points).** Continue to use the model and notation from part (b), and treat p as a fixed candidate report.

(i) One class (6 points). Use the mean and variance from part (b) to show that

$$E[(p_k - \hat{q}_k)^2 \mid x] = (p_k - q_k)^2 + \frac{q_k(1-q_k)}{m}.$$

(ii) Sum and compare (5 points). Sum over the three classes. Then compare the result with part (a), where the Brier score was computed against one random categorical label. In 1–2 sentences, explain what the factor $1/m$ tells you.

(iii) Minimize and take a limit (4 points). Find the candidate report p that minimizes the expected Brier score for any positive integer m. Then find the limit of the expected score as $m \to \infty$ and explain it in one sentence.

Problem 2 — A benchmark label is constructed, not found (30 points)

Use the 72 items in `homework_items.csv`. Work at the **sentence level**: five rater rows are repeated measurements of one item, not five independent evaluation cases. The majority label is an aggregation rule applied to those measurements; it is not an additional observation.

a. **Where disagreement occurs (6 points).** Describe the distribution of vote patterns overall and separately for Rounds 1 and 2. Include one table with the item count, round total, and within-round proportion for every round-by-pattern combination, plus one figure showing the same comparison. In 1–2 sentences, describe the main pattern. Keep every denominator visible.

b. **Performance by vote pattern (7 points).** For `5-0`, `4-1`, and `3-2` items, report the number of items, number correctly classified under the majority label, and majority-label accuracy. In 2–3 sentences, describe the comparison and explain why rater disagreement is not automatically "bad data." Do not add confidence intervals; most interval construction comes in Week 4.

c. **Rounds are construction processes, not treatments (7 points).** Compare Round 1 and Round 2 using majority-label accuracy. Report the item count, number correct, accuracy, and Round 2 minus Round 1 difference. Then write 3–4 sentences that:

1. state what the assigned data show;
2. use one specific detail from the assigned DynaSent documentation to explain how the rounds were constructed;
3. give two plausible explanations for the observed difference; and
4. explain why the comparison does not identify a causal effect of collection round.

d. **One planned sensitivity analysis (5 points).** Before computing the alternative result, choose and record **one** of these changes:

- keep all items and replace majority-label accuracy with mean Brier score against the rater proportions; or
- keep majority-label accuracy and exclude all `3-2` items.

Complete the starter comparison table with each analysis's outcome, retained population, round estimates, and Round 2 minus Round 1 difference. In 2–3 sentences, state what changed and whether the substantive comparison survived.

e. **Red-team the leaderboard with AI (5 points).** Attempting this part is required to earn these 5 points; successfully increasing accuracy is not required. Tell an AI assistant that its job is to make the fixed model's reported majority-label accuracy look better **without changing the model**. Ask it for one transparent, superficially defensible rule

that filters items using only sentence text or rater-disagreement information. The rule may not use model probabilities, model predictions, correctness, or the result of part (d), and you may not revise it after seeing the answer.

Preserve the initial prompt, state the rule and predicted direction before computing, then apply it once. Report the item count, number correct, and accuracy before and after filtering. In 1–2 sentences, explain what changed, why the model itself did not improve, and why this would be a questionable way to build a benchmark. Connect your explanation to the throughline: the selection rule changed which measured items became the evaluation target.

Problem 3 — Three forecasters, one measured target (20 points)

Use the ten purposefully selected DynaSent test sentences in `forecaster_items.csv`. These cases were selected to make disagreement visible; they are a demonstration, not a probability sample. The five DynaSent labels are the measurements. You and the two frozen classifiers are **forecasters** of a target constructed from those measurements:

- `cardiffnlp/twitter-roberta-base-sentiment-latest`, revision `3216a57f`;
- `lxyuan/distilbert-base-multilingual-cased-sentiments-student`, revision `cf991100`.

The starter notebook downloads the public checkpoints and runs them locally; no paid API is required. Do not inspect the five-rater vote columns before completing part (a).

a. **Be the first forecaster (4 points).** Before running either model, report your own probability vector over `negative`, `neutral`, and `positive` for every sentence. Each row must be nonnegative and sum to one. Add your highest- probability label and identify the two sentences for which you were least certain, with one short text-based reason for each.

b. **Run two frozen models (5 points).** Run the supplied checkpoint cell and preserve the raw probability output. Complete the supplied ten-row table with both models' three probabilities, highest-probability labels, and a `models_disagree` indicator. Show that both probability vectors sum to one for every row and report the number of disagreements out of ten. Do not interpret that fraction as a population estimate.

c. **Compare two constructed targets (5 points).** Reveal the vote counts and construct both $\hat{q}$, the three rater proportions, and Y_{maj}, its one-hot majority-vote version. For you and each model, report the mean Brier score against **each** target and the number of highest-probability labels matching the majority; keep the denominator of ten visible. Briefly explain what information is lost when $\hat{q}$ is replaced by Y_{maj} and why neither target becomes literal ground truth.

d. **Measurement check and synthesis (4 points).** Select one case where the models disagree and quote only the shortest phrase needed to identify its ambiguity. For that case, report $\hat{q}$, Y_{maj}, and the three forecasters' highest-probability labels. Explain what the soft target records that the majority target discards. Use one specific fact from each linked model card as a plausible—not proven—explanation for the differing forecasts.

   Finally, organize the data as one row per sampled sentence X_j, with the student and two model forecasts in three aligned columns. Explain (i) under what sampling conditions the sentence-level rows could be iid, (ii) why the three forecasts within a row are not justified as independent, identically distributed rater draws, and (iii) why the student and two models are not exchangeable. Your explanation should use the fact that all three columns respond to the same random sentence, while fixed model outputs are deterministic conditional on $X_j = x_j$.

Required AI-use record (2 points)

Complete the AI-use table in the starter notebook for Problems 2(e) and 3. For Problem 2(e), record the **initial prompt only** rather than the whole conversation. For Problem 3, record both checkpoint IDs and revisions; write "no natural-language prompt" for the prompt field. For both uses, record the purpose, what you checked, what changed, and one decision you retained. Preserve Problem 3's raw probabilities separately.